SPEED DYNAMICS ANALYSIS IN BHARATANATYAM ADAVUS

UIT2718 PROJECT WORK PHASE I

PROJECT REPORT

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BONAFIDE CERTIFICATE

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Certified further that, to the best of our knowledge, the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Bharatanatyam Adavus consist of structured rhythmic movement patterns in which execution speed, the timing of repeated foot actions and consistency across repetitions contribute to the organization of a performance. Conventional study of these aspects depends substantially on observation by experienced teachers or dancers. This project develops a video-based computational approach for examining temporal characteristics of Adavu footwork using pose estimation, interval-based feature extraction and statistical analysis.

The implemented workflow reads a performance video, extracts its frames and applies an RTMPose WholeBody model to estimate foot landmarks. Although six foot landmarks are recorded, the temporal analysis uses the vertical coordinates of the left big toe, left heel, right big toe and right heel. Local minima of each vertical-coordinate signal are detected using SciPy with specified distance, prominence and coordinate criteria. Consecutive detected events define elapsed-frame intervals, which are merged across the four landmarks to produce a one-dimensional feature dataset. K-Means clustering with three clusters groups intervals into relative fast, medium and slow categories by ordering their centroids. Separate Gaussian descriptive analyses characterize the mean, variance, standard deviation and observed standard-deviation coverage of the resulting interval clusters.

The pipeline accepts multiple videos and stores results independently for each recording. For the present Phase I work, the procedure was run on a collection of 56 video performances. The supplied S01 result set is presented as a detailed documented example. It contains 1,788 frame records and 70 pooled inter-event intervals; the three K-Means centroids are 40.17, 115.31 and 225.50 frames. These outcomes demonstrate an interpretable separation of relative inter-event timing. They do not by themselves establish a quantitatively validated classification accuracy or a direct measurement of movement difficulty. The work provides a reproducible basis for examining movement-derived rhythm and for future investigation of other Adavus and repeated performances.

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Equation 3.1: Vertical-coordinate time series

Notation: $y_k(i)$: vertical pixel coordinate; k: foot landmark; i: frame index.

Equation 3.2: Local peak frame intervals

Notation: $\Delta f_k,j$: gap in frames; f_k,j: frame of the jth peak; m_k: peaks for landmark k.

Equation 3.3: Elapsed time in seconds

Notation: $\Delta t_k,j$: elapsed time (s); $\Delta f_k,j$: frame interval; FPS_source: source-video frame rate.

Equation 3.4: K-Means within-cluster objective

Notation: J: objective function; Cc: cluster c; xi: observed interval; μc: cluster centroid; c: cluster index.

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Notation: μfast, μmedium, μslow: cluster mean intervals; Δf: representative frame interval.

Equation 3.6: Sample mean

Notation: μ: sample mean; n: number of intervals; xi: ith interval.

Equation 3.7: Sample standard deviation

Notation: s: sample standard deviation; n: number of intervals; xi: ith interval; μ: sample mean.

Equation 3.8: Gaussian probability-density function

Notation: p(x; μ, s): Gaussian probability density; x: frame interval; μ: mean; s: sample SD; π: pi.

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Notation: Coverage(k): percentage within k SDs; N_k: observations within k SDs; n: sample size.

Equation 4.1: Relative frame-interval frequency

Notation: FPS_source: source frame rate; μinterval: mean observed frame interval (frames).

LIST OF ABBREVIATIONS

AI – Artificial Intelligence

BPM – Beats per Minute

CPU – Central Processing Unit

CSV – Comma-Separated Values

FPS – Frames per Second

ML – Machine Learning

ONNX – Open Neural Network Exchange

RGB – Red Green Blue

RTMPose – Real-Time Multi-person Pose Estimation model family

SD – Standard Deviation

CHAPTER 1
INTRODUCTION

1.1 Overview

Bharatanatyam is a classical Indian dance tradition distinguished by codified body positions, coordinated footwork, rhythmic patterns, expressive gestures and structured combinations of movement. Adavus form an important practical foundation for learning and refining these patterns. A single Adavu comprises a recognizable movement sequence that can be repeated or executed at different rhythmic speeds. Movement technique and artistic expression remain central to its interpretation, while the measurable timing of successive actions offers a useful quantitative perspective.

The present project studies speed dynamics in Bharatanatyam Adavus using recorded video. Rather than treating the video as a collection of still postures to be classified, it treats changes in foot-landmark coordinates through time as an observable signal. Vertical movement patterns are recorded frame by frame; selected recurrent events are located in these signals; and intervals between those events become the basic unit for statistical examination. In this manner, a high-dimensional moving image is reduced to a physically interpretable temporal measurement: the number of frames separating consecutive detected foot-motion events.

The current implementation focuses on Tattadavu. Its repeated lifting and stamping actions provide a relevant setting for measuring periodic foot movement and comparing slower and faster portions of a recording. Three performance-speed categories are used in the analysis. A key objective is not merely to assign numerical categories, but to obtain results that can be traced back to observable vertical foot motion in the original video and to intermediate peak graphs.

The core processing components are frame extraction, RTMPose WholeBody landmark estimation, SciPy peak detection, one-dimensional K-Means clustering and Gaussian descriptive analysis. The complete procedure processes the video files independently and saves their intermediate and final outputs using a consistent directory structure. It can therefore operate on a variable number of video recordings. A collection of 56 performances was

processed for the present study; the S01 outputs supplied with the project documentation permit detailed case-level analysis.

This chapter introduces the motivation, background and boundaries of the work. Later chapters place it in the context of computational dance research, explain the actual implementation and discuss what the available experimental data support.

1.2 Motivation

Dance education depends extensively on the demonstration of a movement by an instructor and the learner's attempt to reproduce its sequence, timing and coordination. A knowledgeable observer can interpret artistic and technical subtleties that no single numerical feature can capture. Nevertheless, when a performance includes many repeated actions and several changes of speed, a quantitative record can complement visual teaching by identifying interval patterns that are not easy to describe consistently from memory.

Adavu performances are particularly suitable for event-based analysis because many training sequences contain recurrent lower-limb movements. The temporal distance between repeated observed movement events can be related to the speed of their execution. When the same event definition is used consistently, larger intervals indicate less frequent repetition and shorter intervals indicate more frequent repetition. This approach creates a practical path from video observations to numerical speed categories without requiring specialized ground-reaction instruments or wearable sensors.

Another motivation is systematic digital documentation. A video preserves the appearance of a performance but does not automatically describe which motion features recur, how the spacing of events changes, or whether one region of the sequence appears more temporally variable than another. Converting the recorded foot motions into a landmark dataset and a series of event intervals makes the performance amenable to repeatable computational analysis.

The current system also demonstrates the value of using existing machine-learning and statistical tools for a culturally specific movement domain. RTMPose supplies landmark coordinates from ordinary video; SciPy identifies candidate movement events; K-Means groups intervals without requiring manually assigned speed labels; and Gaussian statistics summarize the spread within those groups. Each stage is kept interpretable and produces intermediate files that can be inspected separately. This traceability is an important design

feature for a project in which the reliability of the result depends on the reliability of several preceding signal-processing stages.

1.3 Background

1.3.1 Adavus and the role of rhythm

An Adavu is a structured unit of Bharatanatyam training involving coordinated steps and postural patterns. Repetition supports the acquisition of technique and rhythmic control. Traditional instruction often practices movement sequences at several speed levels. This project adopts the slow, medium and fast interpretation supplied with the recorded Tattadavu study rather than claiming that every Adavu or school uses identical temporal ratios.

Tattadavu emphasizes repeated foot actions. From a temporal analysis perspective, the key observable quantity is not an abstract subjective impression of speed but the separation between comparable recurrent movement events. That quantity becomes especially useful when a single recording contains multiple speed portions, because the intervals are expected to change during the progression.

1.3.2 Computer vision and pose estimation

Computer vision methods can transform a sequence of video frames into structured observations. Whole-body pose estimation is a relevant technique when the required measurements correspond to identifiable body landmarks. In the implemented software, the RTMPose WholeBody model produces keypoint coordinates and confidence values for detected persons in each frame. The program focuses on foot landmarks because the input recordings and the research question concern foot movement, not full-body gesture recognition.

The underlying coordinate system describes locations within each image. For a conventional image, the origin is at the upper-left corner, the horizontal coordinate increases toward the right, and the vertical coordinate increases downward. Accordingly, upward motion of a keypoint often appears as a reduction in its Y value. Detecting minima in this signal is thus a plausible means of identifying candidate lifted-foot events; it does not prove exact floor-contact timing.

1.3.3 Event intervals and statistical grouping

A temporal event is a selected location in a measured signal, here the local minimum of a landmark's vertical coordinate under configured constraints. Two successive events define an inter-event interval. Repeating this extraction creates a sequence of elapsed-frame measurements. Each interval retains information about local repetition rate and may be converted into seconds when the original recording frame rate is known.

K-Means is an unsupervised clustering method that groups numeric observations around centers while reducing within-group squared deviations. Its use is appropriate for discovering relative groups of short, intermediate and long intervals when the desired number of groups has been fixed. In this work the value of K is three, corresponding to the three intended speed categories. Gaussian statistical summaries are then computed within the discovered groups to characterize variation; this is a descriptive stage rather than another classifier.

1.4 Problem Definition

The central problem is to extract reproducible, interpretable measurements of execution-speed variation from Bharatanatyam Adavu video without relying on manual frame-by-frame annotation for every event. A performance video contains image appearance, posture changes, lighting, camera viewpoint, movement and possible pose-estimation uncertainty. The analysis must isolate a meaningful temporal signal while retaining sufficient detail to compare recurring actions.

For the present implementation, the input is a recording containing the feet during a Tattadavu performance. The computational task is to estimate the positions of selected foot landmarks across frames, detect candidate recurring vertical-motion events, calculate intervals between them, and partition these intervals into three relative speed categories. A secondary objective is to describe the amount of timing variation within the resulting categories using mean, variance, standard deviation and empirically observed coverage around the mean.

Several boundaries are essential. The method classifies numeric intervals, not whole Adavu types. The selected Y-coordinate minima need not coincide exactly with the physical contact of a foot against the floor. The speed labels derive from ordered K-Means centers rather than supervised ground-truth training. Gaussian spread expresses timing variability within a cluster; its direct

interpretation as movement difficulty is not established by the supplied experiment. These limitations are explicitly retained so that the findings do not exceed what the outputs demonstrate.

1.5 Objectives and Scope

The wider project objectives are to collect and organize representative Bharatanatyam Adavu performances, examine characteristic timing patterns, create quantitative records of speed variation, and establish a foundation for subsequent intra-Adavu and inter-Adavu comparison. The current completed implementation contributes to those objectives through a modular processing pipeline and a documented set of per-video outputs.

The technical objectives for the present phase are: (i) extract the frames of each video; (ii) estimate and store foot-landmark coordinates; (iii) obtain vertical movement signals for four selected landmarks; (iv) identify local minima and inter-event intervals; (v) consolidate interval data across landmarks; (vi) group interval observations into three relative speed categories; and (vii) quantify each category's central tendency and dispersion using Gaussian descriptive analysis.

The software is not restricted to a particular number of recordings. Detailed original outputs are available in the supplied S01 archive, while the remaining recordings and their output folders reside in the project's local environment. Consequently, the report distinguishes system-wide processing scope from directly inspected case-level numerical findings.

1.6 Organisation of the Report

Chapter 1 introduces the topic, motivations and research objectives. Chapter 2 discusses selected prior work on Bharatanatyam analysis, dance tempo estimation, motion alignment, beat annotation and pose-aware methods. Chapter 3 describes the implemented video-to-statistics processing architecture in enough detail to trace each computational output to its input. Chapter 4 documents the experimental arrangement and examines the original S01 outputs, including peak counts, interval distributions, clustering and Gaussian statistics. Chapter 5 considers educational, cultural and sustainability implications. Chapter 6 summarizes the work and its supported findings, and Chapter 7 identifies appropriate directions for future improvement. The references and an appendix complete the report.

1.7 Functional Requirements and Working Assumptions

A functional requirement of the system is that the input recordings can be decoded into ordered frames. The user should not need to perform the same landmark extraction and statistical steps manually for every recording. Instead, the processing sequence should locate its input video, create a corresponding output directory, execute the components in the required order and persist their results. This requirement is reflected in the master controller and the separate functions in each module. A related requirement is traceability: the final statistical graphs must be supported by saved intermediate measurements that can be inspected when the output needs to be explained.

The foot-specific requirement is to obtain a coordinate time series for each landmark used in analysis. The system records six foot landmarks but focuses its temporal calculations on the two big toes and two heels. This approach is intentionally more limited than a complete whole-body motion interpretation. It reduces the feature-extraction problem to signals that are relevant to foot lifting and lowering. Maintaining landmark identifiers ensures that results can later be related back to the correct foot and anatomical location, even after interval data are pooled for clustering.

The statistical requirement is to distinguish relative groups of interval duration, not to infer artistic quality automatically. The number of requested groups is fixed at three, corresponding to the assumed three speed portions of the documented performance. Since the model operates on interval length, each valid observation must be a numeric elapsed-frame value derived from two successive accepted events. The clustering output should preserve the input interval and landmark identifier together with its assigned group, allowing the viewer to audit the transformation.

Several working assumptions affect interpretation. The video is presumed to show enough of the relevant foot movement for RTMPose to estimate landmarks meaningfully. The camera viewpoint and crop should be sufficiently stable that fluctuations in Y coordinates are predominantly associated with foot movement rather than global camera motion. The selected event type is a local vertical-coordinate minimum, and repeated selected events are treated as comparable for the purpose of interval analysis. These assumptions are practical choices in the current system; the supplied output material does not quantify how often each assumption holds across all recordings.

The intended software is exploratory and research-oriented. It produces measurements and plots that are suitable for further examination by a researcher or dance expert; it is not designed to issue an authoritative pass/fail evaluation of a dancer. A robust future application would need to communicate event-detection uncertainty, accept recording metadata, and identify situations in which the input video cannot support a reliable inference. Stating these boundaries in the requirements helps protect against interpreting automatically produced numbers as complete descriptions of Bharatanatyam performance.

CHAPTER 2
LITERATURE SURVEY

The selected literature is organized around six complementary directions: force-based Adavu speed classification, inertial dance-tempo estimation, audio beat annotation, Kinect-based tutoring, posture and sequence recognition, and pose-aware generation of dance postures. These publications motivate the relevance of quantitative movement analysis but also show that speed measurement, motion recognition and artistic quality assessment are distinct tasks. The discussion below separates each publication's reported contribution from the specific video-based implementation in this project.

2.1 Adavu Speed Classification Using Force-Platform Data

Adalarasu and co-authors (2025) investigate classification of Bharatanatyam Adavu execution speed using signals acquired from a force platform. The experimental approach measures biomechanical interactions that arise as the feet act on the ground. Ground-reaction-force and center-of-pressure information can capture features that are directly linked to support, weight transfer and the timing of force generation. These measurements provide a quantitative alternative to purely visual judging of repeated dance steps.

The study's methodological significance lies in treating Adavu speed as a machine-learning classification target. Features extracted from recorded force-platform signals are supplied to supervised classifiers, allowing the authors to evaluate the separability of speed categories. As described in the research literature and the existing project survey, this is a direct precedent for computational analysis of Bharatanatyam tempo. Importantly, the published experimental speed setup should not be conflated with the three-cluster configuration of the present project; the two works are not identical in data representation, labels or validation approach.

A force platform is advantageous when actual contact forces and loading changes are required. It also introduces a dependence on specialist hardware and a controlled measurement environment. An ordinary camera recording does not provide ground-reaction force, but it may preserve enough recurrent motion information to estimate intervals between candidate events. The present implementation therefore operates in a different signal domain: vertical image coordinates rather than measured forces.

The distinction matters for comparison. The published classifier results cannot be treated as benchmarks achieved by the video-based method, and the methods cannot be compared numerically without matched recordings, equivalent labels and consistent evaluation criteria. The principal relevance is conceptual: Adavu speed can be made into an explicit quantitative analysis target. A future matched study could examine how well peaks found in vertical foot motion align with independently measured force-contact events.

2.2 Dance Tempo Estimation from a Leg-Mounted Accelerometer

Stančin and Tomažič (2021) present a wearable-sensor approach to estimating dance tempo from three-dimensional acceleration measured on a dancer's leg. The study considers solo-jazz movement sequences containing different activation patterns and rhythmic variations. A bank of enhanced multiple resonators implemented through comb-feedback filters is used to recover periodic behavior in the acceleration signals.

The authors evaluated fifteen distinct solo-jazz moves at eight metronome tempos covering 80 to 220 beats per minute. Their reported tempo errors were at most one beat per minute across measurement instances in the studied setup. The paper further discusses shorter movement excerpts, demonstrating that tempo can sometimes be inferred without processing an entire long performance. These results are specific to the wearable-sensor arrangement and the described dance recordings (Stančin and Tomažič, 2021).

The study supports the general idea that cyclical movement of the lower limbs contains information about intended rhythmic tempo. Acceleration emphasizes changes in velocity, whereas the present project uses image-space landmark position directly. A video-based coordinate signal can be tracked retrospectively from a recorded performance and does not require a device physically attached to the dancer. In exchange, it may be sensitive to occlusion, cropping, background ambiguity and camera changes.

The two methods also differ in the type of output. Resonator-based analysis estimates a tempo related to repeated motion; the present system groups measured frame intervals into three relative speed categories. No claim of beats-per-minute accuracy is made here. The principal lesson for the project is methodological: repetitive leg motion is an observable carrier of temporal structure, and extracting that structure may assist dance analysis.

2.3 Beat Detection and Annotation of Bharatanatyam Sollukattu

Mallick, Das and Majumdar (2020) examine rhythmic information in the audio accompaniment of Bharatanatyam. The system analyzes Sollukattu, which consists of structured rhythmic utterances and instrumental beats. Speech-recognition methods are applied to vocalized syllables, and the structure of the recognized patterns is used with beat-related measurements to support automatic annotation of the music.

The authors report bol-recognition accuracy of approximately 85 percent, Sollukattu-recognition accuracy of 95 percent, tempo-period estimation accuracy of 96 percent and beat-annotation performance above 90 percent in their reported experimental setting. These figures refer to the audio processing described by Mallick and colleagues and are not results obtained by the current video-analysis pipeline.

This publication is relevant because it establishes a different reference for rhythmic timing: the temporal organization of the accompanying sound. A dancer's visually observed foot motion and the musical beat may be related, but they are not interchangeable measurements. Small timing offsets can arise between foot lifting, the highest motion position, the downward movement and ground contact. An audio-side beat timeline could help determine which of these visually detected events best corresponds to the intended dance beat.

The present work deliberately concentrates on movement-derived interval patterns and does not calculate an audio beat signal. The relationship between audio rhythm and visual movement is thus a possible validation path rather than an implemented subsystem. A well-controlled future experiment could annotate musical beat timestamps and compare them to peaks, minima or other contact-related features extracted from foot trajectories.

2.4 NrityaGuru: Kinect-Based Dance Tutoring

Aich and collaborators (2018) describe NrityaGuru, a Bharatanatyam tutoring approach using Kinect-based movement capture. A reference performance can be compared with a learner's recording, and differences in body movement or posture can be identified. Because two performances may differ in duration and execution speed, direct frame-to-frame comparison can be misleading.

Dynamic Time Warping is relevant in this context because it creates an alignment between sequences that unfold at different rates. Rather than

assuming frame i of one recording corresponds to frame i of another, an alignment method finds a flexible correspondence that accommodates local stretching or compression in time. The reported tutoring framework uses this general idea to compare recorded motion across distinct executions.

The present project addresses a related but narrower problem. Its immediate objective is to obtain event intervals and characterize speed categories within an individual video. It does not contain a reference-dancer matching module or calculate Dynamic Time Warping distances between recordings. Nevertheless, the current extraction of frame-indexed events could provide a compact input for later cross-performance alignment.

The literature therefore helps identify a future extension: after local event intervals have been measured reliably, repeated recordings of the same Adavu could be aligned and compared to distinguish genuine rhythmic changes from differences in recording length. Such a step would require carefully chosen correspondence rules and evaluation data; it is not assumed to be part of the current verified results.

2.5 Posture and Sequence Recognition in Bharatanatyam

Mallick, Das and Majumdar (2022) study recognition of action elements in Bharatanatyam performances. Their published paper in the Journal of Visual Communication and Image Representation addresses segmentation into representative dance elements, identification of postures and recognition of ordered sequences. The work combines music-related information and movement observations to exploit the structured nature of Adavu performance.

The investigated dataset includes Natta Adavu material with representative key postures. The approach reported in the available paper material applies feature-based and neural methods to posture recognition and sequence-modeling methods to determine the structure formed by successive action elements. The study provides evidence that recurring postural units and the order between them are useful building blocks for computational dance representation.

A posture-recognition system answers the question of which configuration or dance element appears at a particular instant. The present system asks a different question: what is the frame interval between consecutive selected vertical foot-movement events, and how are these intervals distributed? These tasks are related through their shared concern with temporal structure but are

not equivalent. Recognition accuracy reported for postures must not be interpreted as evidence of speed-clustering accuracy.

The conceptual connection is the choice of a discrete event as the bridge between continuous motion and interpretable dance structure. The present pipeline selects coordinate minima as candidate events; posture-oriented methods select representative configurations. Future work could investigate whether combining the two produces more reliable, dance-aware event detection than a coordinate-only threshold.

2.6 Generating Key Postures with Pose Estimation

Kamble, Mukhopadhyay, Roy and Das (2026 preprint; connected to ICVGIP 2025) investigate the generation of Bharatanatyam key postures using pose-aware generative modeling. Their work examines how pose-estimation supervision and pose-consistency constraints can support the production of movement configurations that better respect recognizable anatomical and stylistic structure. The emphasis is on generating or representing valid key postures rather than measuring event-to-event speed intervals.

This line of research is relevant to the broader objective of digitally preserving and analyzing structured dance vocabulary. It illustrates that landmark-based information can be incorporated into computational methods for Bharatanatyam, and that domain-specific representations matter when evaluating the realism of generated or reconstructed movement.

The present project neither synthesizes postures nor uses a generative model. Its reliance on detected foot landmarks is simpler, and the immediate output is a numeric distribution of timing features. The cited study is therefore best treated as background on modern pose-aware Bharatanatyam computing, not as a directly comparable speed-classification baseline.

A future study could examine whether more detailed pose-quality measures improve the rejection of unreliable landmark detections. In the implemented pipeline, confidence values are recorded but the downstream timing analysis chiefly uses four coordinate columns. The distinction between merely detecting a coordinate and establishing a stable, anatomically plausible landmark trajectory deserves systematic evaluation.

2.7 Comparative Analysis of Existing Work

Table 2.1: Comparison of the selected related studies

Study	Primary data	Central task	Difference from present work
Adalarasu et al. (2025)	Force platform	Adavu speed classification	Video coordinates and unsupervised interval groups
Stančin and Tomažič (2021)	Leg accelerometer	Dance tempo estimation	No wearable; frame-event intervals
Mallick et al. (2020)	Sollukattu audio	Bol, beat and tempo annotation	Motion signal, not audio timing
Aich et al. (2018)	Kinect capture	Tutoring and temporal alignment	Single-video timing statistics
Mallick et al. (2022)	Multimodal dance video	Posture and sequence recognition	Foot-event interval grouping
Kamble et al. (2026)	Pose-aware dance corpus	Key-posture generation	Analysis without posture synthesis

The comparison reveals two common themes. First, structured dance actions can be studied computationally when suitable measurements of motion or rhythmic events are available. Second, the measurement source significantly affects what can be claimed: force plates describe loading, accelerometers describe inertial motion, audio describes beat organization, and images describe observable position changes.

The proposed work is situated at the intersection of movement-derived timing and video pose estimation. It provides a practical pipeline that produces separate outputs for each recording and expresses speed in relative inter-event interval categories. Its immediate contribution is implementation and documentation of this analysis strategy for the available performances; a broader claim of novelty against every published RGB-video method would require a dedicated systematic literature search beyond the six selected studies.

2.8 Research Gap and Positioning of the Present Work

The reviewed studies motivate an opportunity to examine repeated Bharatanatyam foot actions using ordinary recorded video, explicit foot-landmark traces and a simple interval-based statistical representation. This

approach complements methods focused on force, wearable acceleration, musical beats, key-posture recognition or generative pose modelling. It is particularly suitable for obtaining interpretable, per-video graphs and interval tables from recorded material.

The research gap addressed here is operational rather than a claim that no video-based dance timing study exists: a reproducible workflow is needed to transform foot-specific pose trajectories into candidate motion events, interval data and comparable descriptive summaries. The present implementation supplies that workflow, demonstrates three relative interval groups in S01 and records Gaussian statistics for each group.

The next stage of research should validate detected motion events against expert or contact-ground-truth annotations, evaluate consistency across dancers and recording setups, and test whether cluster labels correspond reliably to known speed segments. Such validation is required before the approach can be described as an accurate speed-assessment instrument instead of an exploratory quantitative analysis pipeline.

2.9 Implications for Experimental Design and Interpretation

The prior literature helps distinguish several ways in which the word “speed” may be measured. In music, tempo can be expressed as a rate of beats per minute; in biomechanics, movement frequency may be inferred from force or inertial signals; in image-based analysis, a rate can be derived from intervals between repeatedly detected positions or events. These measurements may be correlated in a well-executed dance sequence, but none is inherently identical to the others. An experimental report must therefore state the observation that defines each event and the units in which its timing is measured.

The present system uses the number of elapsed frames between successive minima in a landmark’s vertical coordinate. It does not measure force, the downward speed of contact, the loudness of stamping, or musical beat onset. The units are frames until a recording frame rate is used for time conversion. For example, a difference of 40 frames carries a different absolute duration in recordings acquired at 25 and 30 frames per second. This is not a contradiction in the clustering algorithm; it reflects the difference between a frame-based feature and a time-normalized feature.

Work employing supervised classification, such as force-platform speed analysis, requires an external labelling process. The model can then be assessed

by comparing predicted labels with the intended labels. In contrast, the present three-cluster K-Means implementation has no ground-truth labels at training time. The model organizes its input intervals by similarity and the labels Fast, Medium and Slow are assigned after the fact by sorting cluster means. The existence of three resulting groups is guaranteed by the selected value of K when enough distinct observations are present; therefore, three visible groups cannot alone prove that a video contains three correctly recovered traditional speed stages.

Research on audio beat annotation offers another independent observation channel. If a rhythm track and a performance recording are synchronized, the temporal difference between detected motion events and annotated beats may be examined. Such an approach would support studies of entrainment and coordination. However, connecting two signals requires a consistent time base, careful handling of systematic delays and an understanding of which phase of the movement corresponds to the beat. A lifted toe, a lowered heel and a contact impulse are not necessarily simultaneous events.

Posture and sequence recognition studies highlight yet another distinction: recognition of what movement configuration is present is different from analysis of how frequently a repeated configuration occurs. A high-performing classifier that recognizes a particular posture would not automatically determine the speed of successive repetitions. Conversely, grouping event intervals into three categories would not identify the underlying Adavu type from an unknown collection of performances. Both tasks could eventually be combined, but the computational aims and evaluation criteria must remain separate.

The literature also suggests how to structure fair comparisons. The recording population, range of speeds, data modality, number of performers, event definition, labelling process and train/test design would need to be held sufficiently comparable. Without these conditions, a reported accuracy from another paper is not a measured baseline for this method. A strong research contribution can still be established by demonstrating a transparent and reproducible processing method, identifying the exact variables measured, and describing the additional experimental controls needed before drawing stronger performance conclusions.

For this Phase I report, the selected six papers serve as a grounded research context rather than as an exhaustive survey of every publication on computational dance. Their methods motivate movement-derived timing

analysis and suggest validation paths, but they cannot establish a universal negative claim such as the absence of any previous RGB-video tempo study. The operational contribution is a complete implemented pipeline with inspectable intermediate and final artifacts, applied to the available 56-video dataset and documented in detail for the supplied S01 performance.

CHAPTER 3
PROPOSED METHODOLOGY

3.1 System Overview and Processing Architecture

The architecture of the developed system is a sequence of modular transformations. The input is a digital video of an Adavu performance. The first module reads and stores individual frames. The second estimates foot landmarks on each frame and creates a structured landmark table. The third converts vertical trajectories into event markers using peak-detection rules. The fourth creates a pooled dataset of consecutive event intervals from four selected foot landmarks. The fifth groups those numeric intervals into three speed categories using one-dimensional K-Means. The final implemented analysis module computes descriptive Gaussian statistics separately for each category.

The overall program has a video-independent processing design. Each recording is assigned its own directory, so intermediate outputs and results remain separate. This is important for dataset management and prevents the results of one performance from being combined with another unintentionally. A master controller discovers MP4 files in the input directory and submits processing jobs to a process pool. The supplied configuration uses four worker processes, a value that controls concurrent execution rather than the total permissible dataset size.

Figure 3.1 reproduces the architecture diagram from the supplied project review presentation. The detailed implementation in the remainder of this chapter reflects the current code and focuses exclusively on the movement, clustering and Gaussian-analysis stages.

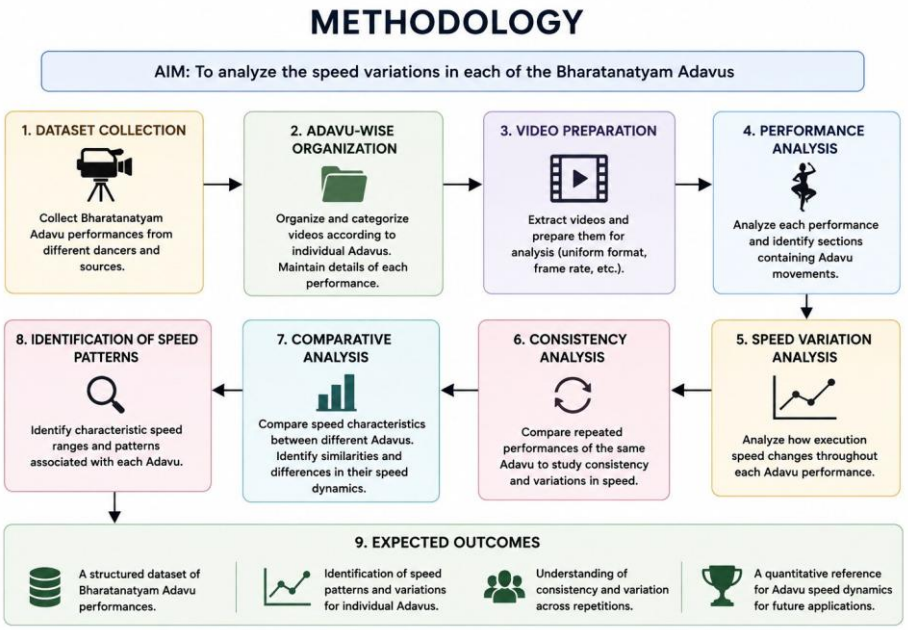


Figure 3.1: Proposed project architecture

3.2 Input Video Dataset and Organization

The software accepts recorded MP4 videos stored within an input directory. For the present phase, the available experimental collection comprised 56 video performances, and the same processing procedure was applied independently to those recordings. This is an experimental dataset count, not a hard-coded system limit. The implemented controller obtains filenames dynamically and therefore permits additional input videos without a redesign of the analysis stages.

The dataset is intended to support organization by Adavu and by performance. The demonstrated case, S01, is a Tattadavu performance with repeated foot actions and sections that can be interpreted at different relative speeds. The present report does not assert unprovided dancer demographics, recording-device specifications, distribution of different Adavu types among the 56 files, or speed-ground-truth labels. Those details are not available in the reviewed dataset material.

For a reproducible experiment, each recording should retain its original video identifier and the derived output directory. A common output format aids comparison and simplifies inspection. The actual implementation stores extracted frames, RTMPose outputs, detected-peak tables and graphs, pooled data and statistical outputs under a directory named after the video.

3.3 Frame Extraction with OpenCV

The file extract_frames.py opens the configured video using cv2.VideoCapture. It first checks that the video can be opened and reads metadata such as frame count, frame rate, width and height. A loop then calls cap.read() until no further frame is returned. Each successfully decoded image is saved as a JPEG file with an incrementing zero-padded name such as frame_000000.jpg. Consequently, the original chronological frame order is preserved in sorted filenames.

The extraction stage does not down sample the recording by discarding every alternate frame or sampling at fixed intervals. Every successfully decoded frame is written. This matters because all later event separations are initially expressed in units of video frames. If frames were skipped, a recorded interval would need to be adjusted to retain its original temporal meaning.

After extraction, the video handle is released. Failures to open the recording are reported explicitly. The extraction module returns the output directory, and subsequent modules use the same pipeline path configuration. The result is an inspectable representation of the input video as an ordered image sequence.

3.4 RTMPose WholeBody Foot Landmark Estimation

3.4.1 Model initialization

The pose-estimation stage is implemented in rtm_pose.py using rtmlib.Wholebody. The model is initialized in balanced mode, with ONNX Runtime as the backend and CPU execution as configured in the supplied software. The method estimates keypoint coordinates and confidence values from each frame. These values are associated with a detected person in the frame; the implemented extraction uses the first detected person where person detections are available.

Although the input task concerns a feet-focused recording, the model interface used is a whole-body estimator. Its foot-related keypoints are selected using COCO WholeBody indices. The software defines six named foot landmarks and stores horizontal position, vertical position and detection confidence for each. The code also maintains an annotated output video showing the estimated landmark locations where confidence permits their display.

Table 3.1: Foot landmarks used by the RTMPose extraction stage

Landmark index	Model landmark	Used in speed clustering
17	Left big toe	Yes
18	Left small toe	No
19	Left heel	Yes
20	Right big toe	Yes
21	Right small toe	No
22	Right heel	Yes

3.4.2 Landmark coordinates and confidence

For frame i, each detected landmark has an image-plane position represented by X and Y coordinates in pixels. Y is the feature used for temporal motion analysis because repeated lifting and lowering of the feet can produce vertical-coordinate fluctuations. The X coordinate and confidence values are still useful diagnostic measurements and are retained in the raw landmark spreadsheet.

A configured confidence threshold of 0.30 affects the display of detected markers in the visualization stage. The coordinate fields themselves are written into the output table as provided by the model. Therefore, the presence of an estimated coordinate in the spreadsheet should not automatically be interpreted as proof that the underlying landmark was detected with high confidence. More stringent exclusion or interpolation rules could be added in future work, but those are not assumed here.

The pose-extraction script records frame index and nominal time in seconds, alongside the coordinate and confidence fields. Its current FPS constant is 30. The frame-extraction module separately reads the source video's actual FPS. These values should be checked for agreement when interpreting frame intervals as absolute seconds, especially across videos captured at different acquisition rates.

$$y_k(i) := Y_{k,i} \quad (3.1)$$

3.5 Foot Trajectory Construction

The foot trajectory is the sequence of estimated Y coordinates associated with a fixed landmark across successive frames. For each of the six detected foot landmarks, the model stage can create a frame-versus-Y graph. The downstream event-detection stage uses four of these trajectories: left big toe,

left heel, right big toe and right heel. This selection provides information from both feet and from two regions on each foot.

The use of multiple landmarks allows repeated foot motions to be represented from different points, instead of relying on one trajectory that may be less informative during particular phases of a performance. However, different landmarks can mark slightly different instants within one physical foot action; a toe and heel do not necessarily reach their extreme positions simultaneously. The program therefore identifies peaks independently for each landmark before pooling the resulting intervals.

The measured trajectories are expressed in image pixels. These values are not calibrated three-dimensional distances or physical heights above the floor. Differences in camera perspective and person position can influence the amplitude of a trajectory. The current speed feature is based on frame separation rather than the physical displacement amplitude, reducing dependence on pixel scale but not eliminating all effects of pose-estimation noise.

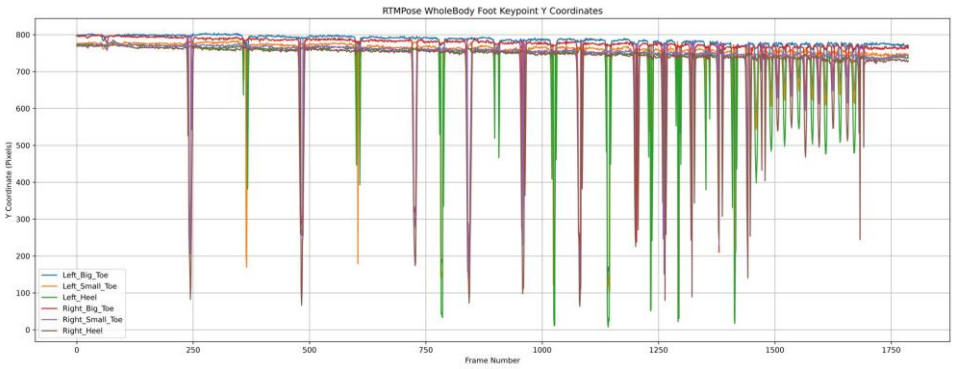


Figure 3.2: Combined vertical-coordinate trajectories for S01

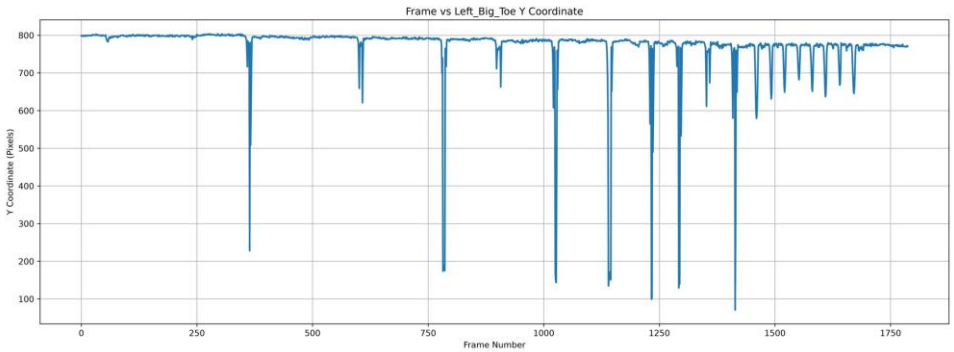


Figure 3.3: Left big toe vertical-coordinate movement

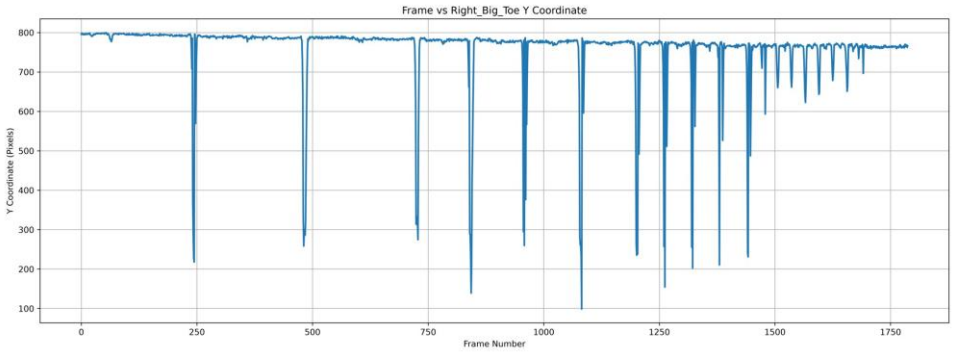


Figure 3.4: Right big toe vertical-coordinate movement

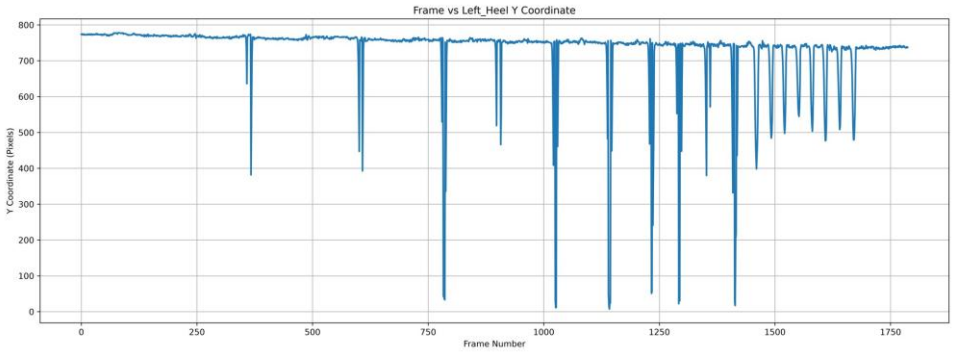


Figure 3.5: Left heel vertical-coordinate movement

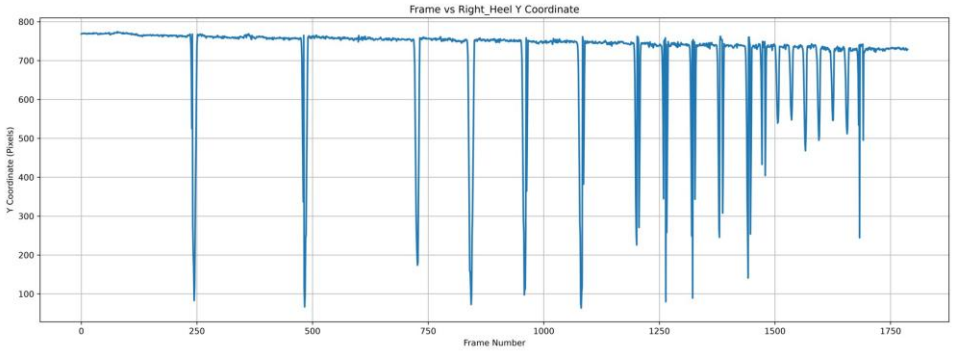


Figure 3.6: Right heel vertical-coordinate movement

3.6 Peak Detection in Vertical Motion Signals

3.6.1 Finding candidate recurrent motion events

The peak_detection.py module reads foot_keypoints.xlsx and the Frame column. For each of the four selected landmarks, it extracts the recorded Y-coordinate series. The code forms the negated signal, -Y, and calls scipy.signal.find_peaks on this series. A maximum of the negated signal is a

minimum of the original Y signal; the selected events are therefore local vertical-coordinate minima under the adopted image-coordinate convention.

The search applies a minimum distance of 20 samples and a prominence threshold of 40. The distance rule prevents candidates from appearing arbitrarily close together, while prominence filters local extrema by how distinctly they rise above the surrounding negated signal. Once these candidates have been found, a separate check retains only those whose original Y coordinate is not greater than 700 pixels. These choices are implementation parameters, not values shown to be optimal for all possible videos.

It is important to distinguish a coordinate minimum from a verified physical stamp. An extreme upward position may precede the downward trajectory that produces floor contact. The project's current event selection is therefore described as vertical movement peak detection or candidate lifted-foot event detection. A floor-contact interpretation would require additional synchronization or annotation.

3.6.2 Peak tables and annotation graphs

For each landmark, the module saves a spreadsheet containing the ordinal peak number, the corresponding frame number, the measured Y coordinate and an elapsed-frame value. The first peak has no preceding peak and consequently has no elapsed-frame measurement. Subsequent rows record the difference between each peak frame and the immediately preceding detected peak frame.

A corresponding annotated graph displays the complete Y trajectory and places red markers at the selected peaks. Each marker is labelled with its frame number. This makes it possible to inspect the local motion pattern and determine whether a detected candidate is plausible rather than relying only on the final clustering results. The design deliberately preserves both a machine-readable interval table and a human-readable visual explanation for every analysed landmark.

$$\Delta f_{k,j} = f_{k,j} - f_{k,j-1}, \quad j = 2, 3, \ldots, m_k \qquad (3.2)$$

$$\Delta t_{k,j} = \frac{\Delta f_{k,j}}{FPS_{source}} \qquad (3.3)$$

3.7 Integrated Interval Dataset

The data_loader.py module reads four detected-peak spreadsheets. From each file it selects the Elapsed_Frames column, drops missing first-peak entries,

converts the retained values to floating-point numbers and adds a Landmark identifier. The four data frames are concatenated row-wise and exported to merged_elapsed_frames.xlsx. The associated clustering feature array is the Elapsed_Frames column reshaped as a two-dimensional NumPy array with one numeric feature per observation.

This dataset is called one-dimensional because every sample presented to K-Means consists only of a scalar interval length. It is not a four-dimensional vector containing four simultaneous landmark values. Landmark identifiers are retained as metadata but are not numerical features used in K-Means fitting. Likewise, the individual peak frame numbers are present in the per-keypoint output spreadsheets, not in the merged one-feature array.

Pooling intervals from multiple landmarks increases the number of available observations in a recording. It also means that the dataset may contain two or more observations associated with different landmarks during one physical repetition. These observations are treated independently by the clustering model. The procedure is intended to discover relative interval sizes, not to identify a unique foot-stamp count or to synchronize the left and right feet at every event.

3.8 One-Dimensional K-Means Speed Clustering

3.8.1 K-Means model

The kmeans.py module reads merged_elapsed_frames.xlsx and constructs the feature matrix X from the Elapsed_Frames column. The scikit-learn KMeans estimator is initialized with n_clusters=3, random_state=42, n_init=20 and max_iter=500. The fitting method assigns each observed interval to a cluster and returns three cluster centroids, each representing the average position of one interval group in frame units.

The K-Means objective is to find a partition that minimizes the sum of squared deviations of points from their assigned centers. This objective explains why intervals of similar numerical length tend to be grouped. K-Means does not read the accompanying dance video to interpret gestures and does not consult a teacher-supplied speed label during fitting; the model's output is an unsupervised partition of scalar intervals.

$$J = \sum_{c=1}^{3} \sum_{x_i \in C_c} (x_i - \mu_c)^2 \quad (3.4)$$

3.8.2 Mapping cluster indices to speeds

Raw cluster numbers are arbitrary, because the index assigned by a clustering algorithm has no inherent semantic meaning. The code sorts the three centroids in ascending order and maps the smallest center to Fast, the middle center to Medium and the largest center to Slow. This mapping follows the feature definition: shorter time intervals correspond to more frequent detected events, whereas longer intervals correspond to less frequent detected events.

After mapping, each original sample receives both a numeric cluster identifier and a readable Speed category. The results are saved as kmeans_results.xlsx. A one-dimensional scatter plot shows the observed intervals along the horizontal frame axis and indicates cluster colors and centroid positions. This visualization is appropriate for auditing the interval grouping, although points may overlap when the same frame difference occurs multiple times.

The labels should be interpreted as relative group labels derived within the processed recording. They are not automatically equivalent to an independently verified Bharatanatyam tempo notation, and the existing software does not transform the cluster labels into explicit start and end timestamps for the three video portions.

$$\mu_{fast} < \mu_{medium} < \mu_{slow} \quad \Rightarrow \quad \Delta f_{fast} < \Delta f_{medium} < \Delta f_{slow} \qquad (3.5)$$

3.9 Gaussian Descriptive Analysis of Speed Groups

The gaussian.py module reads the saved K-Means result spreadsheet and processes Fast, Medium and Slow records separately. It calculates the number of intervals, arithmetic mean, sample variance and sample standard deviation for each group. The module then forms intervals at one, two and three sample standard deviations from the observed mean and counts how many observations fall inside each interval.

These counts are converted to percentages of the observations assigned to the given cluster. A normal probability-density curve is drawn with the cluster's estimated mean and sample standard deviation. The curves provide compact visual summaries of central tendency and dispersion. They do not constitute evidence that a formal goodness-of-fit test has established normality, and the observed coverage percentages need not match the textbook approximate 68–95–99.7 percent associated with an ideal normal distribution.

It is also important that a relatively narrow standard deviation describes concentrated event intervals, not automatically easier dance movement. Timing consistency and movement difficulty are related only through a hypothesis requiring separate validation. The current use of Gaussian statistics is most defensible as a description of intra-cluster temporal variation.

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i \quad (3.6)$$

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \mu)^2} \quad (3.7)$$

$$p(x;\, \mu, s) = \frac{1}{s\sqrt{2\pi}}\, exp\left\{-\frac{(x-\mu)^2}{2s^2}\right\} \quad (3.8)$$

$$Coverage(k) = \frac{100}{n} N_k, \quad k \in \{1, 2, 3\} \quad (3.9)$$

3.10 Software Modules, Outputs and Reproducibility

Table 3.2: Implemented processing modules and principal outputs

Module	Function	Principal output
extract_frames.py	Read video and save individual frames	all_frames/*.jpg
rtm_pose.py	Estimate foot landmarks; annotate motion	foot_keypoints.xlsx, foot_tracking.mp4
peak_detection.py	Detect local Y minima separately	4 peak spreadsheets and 4 plots
data_loader.py	Pool valid inter-event intervals	merged_elapsed_frames.xlsx
kmeans.py	Cluster 1D intervals into three groups	kmeans_results.xlsx, clustering plot
gaussian.py	Compute mean, SD and empirical coverage	gaussian_statistics.xlsx, Gaussian plot
pipeline_paths.py	Create per-video output directories	Dedicated results/<video-id>/ tree
master.py	Discover and dispatch video jobs	Per-video orchestration log

Table 3.3: Principal parameter values

Component	Parameter	Value in supplied code
RTMPose	backend / device	onnxruntime / gpu
RTMPose	mode / display threshold	balanced / 0.30
Peak detection	minimum distance	20 frames
Peak detection	prominence	40
Peak detection	maximum retained Y	700 pixels

K-Means	clusters / random state	3 / 42
K-Means	initializations / max iterations	20 / 500
Orchestration	maximum workers	4
Pose video writer	configured FPS	30

The modular design supports repeatability because parameters and major processing stages are recorded explicitly, and because each stage produces persistent intermediate data. This allows an error in a final plot to be traced backward through the interval spreadsheet, original peak table and raw coordinate data. It also separates model behaviour from experiment metadata: adding more video recordings does not change the meaning of the K-Means input feature or the Gaussian calculations.

The analysis scope of this report is limited to the described peak detection, interval clustering, and Gaussian stages. The described speed-clustering and Gaussian outputs are self-contained preceding stages. The entire experimental result set can be reproduced only where the original input videos, compatible software dependencies and project environment are available.

3.11 Reliability, Failure Cases and Practical Measurement Considerations

3.11.1 Effects of landmark estimation uncertainty

The full analysis depends on the quality of the estimated landmark trajectories. A neural pose estimator may return a numerically plausible coordinate even when a foot is temporarily hidden, the image is blurred by fast motion or the frame contains limited visual context. In a feet-only recording, cropping can alter the visual cues normally available to a whole-body detector. Such effects may produce sudden coordinate jumps, duplicated landmark positions or brief intervals containing missing values. Because the peak detector uses vertical-coordinate extrema, a large spurious displacement can resemble a genuine foot lift unless confidence and temporal continuity are considered.

The code stores the confidence returned for each of the six foot landmarks. This is important diagnostic information. The threshold of 0.30 used in the visualization logic should not be misinterpreted as proof that every coordinate contributing to a final interval exceeds the threshold. The later peak detector reads Y coordinates from the saved spreadsheet and does not independently apply a confidence-based exclusion rule. A future robustness study could

compare event sets obtained with and without confidence filtering, examine missing-value runs and report how many accepted peaks fall in low-confidence regions.

Any attempt to smooth a trajectory must balance two requirements. Noise suppression could remove artificial local extremes, but excessive smoothing may also remove genuine short, sharp movements during faster Tattadavu sections. A suitable validation protocol would start with a manually annotated subset of recordings, select smoothing and threshold parameters on separate development examples, and then evaluate the complete event-detection procedure on held-out videos. Without such a protocol, claiming that a preprocessing choice improves accuracy would be premature.

3.11.2 Peak definition and interval sensitivity

A detection rule may behave differently across slow and fast regions of the same performance. The configured distance parameter of 20 frames prevents accepted extrema from occurring closer than that minimum separation. This can be useful for suppressing duplicate detections on an approximately periodic signal; however, if true relevant events are closer than 20 frames in a particular recording, the rule can cause valid events to be missed. The current S01 Fast cluster has a minimum observed retained interval of 27 frames, which is consistent with the applied distance rule but does not establish that all possible faster events would be retained.

Prominence measures the contrast of a candidate peak against its surroundings in the transformed signal. A prominence value of 40 can exclude minor fluctuations, but image scale and visible movement amplitude influence this criterion. A recording with different cropping or camera placement might yield smaller coordinate displacements even for comparable dance motions. Likewise, the condition Y not greater than 700 pixels assumes a particular numerical region in image coordinates. Since image dimensions vary with resolution and crop, a single absolute pixel boundary may not generalize unchanged across all possible source videos.

The practical implication is that each peak should be regarded as a candidate generated by explicit rules. The saved peak tables and plots permit a reviewer to inspect the candidates rather than accepting them without context. Sensitivity analysis could repeat the method with several plausible parameter combinations and measure the stability of inter-event intervals and the resulting

cluster centres. Such a comparison would be especially useful for recordings with unequal frame rates, greater motion blur or unusual toe and heel visibility.

3.11.3 Data representation and dependence among observations

The pooled input table records one row per valid consecutive event interval for an individual landmark. Because the four landmark streams are processed independently, the observations are not necessarily statistically independent. A physical action involving one foot may generate an interval in the big-toe series and another in the heel series. Both are useful measurements, but treating them as unrelated repetitions could overstate the amount of independent evidence available for estimating population-level uncertainty.

This distinction is particularly important when interpreting standard deviations and any later confidence intervals. The present Gaussian implementation provides descriptive sample variance within cluster membership; it does not use hierarchical models, performer-level random effects, or repeated-measures inference. The sample count of 70 is therefore an interval-observation count rather than a count of 70 different participants, 70 recordings or 70 confirmed independent stamps. For analyses across many videos, a useful future organization would include video identifier, performer identifier where authorized, landmark identifier, event frame pair and speed section annotation.

The current K-Means input also discards time order at the fitting stage. Two intervals of equal duration receive the same numerical feature even if one occurs near the beginning of a video and the other near the end. The original per-landmark peak tables preserve the frame positions and can be used to reconstruct chronological context after clustering. This separation of feature value and sequence position is deliberate in the existing one-dimensional model, and it should be made clear in descriptions of the resulting speed categories.

3.11.4 Reproducibility and model boundaries

For reproducibility, the relevant software components are OpenCV for frame handling, rtmlib with ONNX Runtime for pose estimation, pandas for spreadsheet processing, SciPy for peak detection, NumPy for numerical operations, scikit-learn for K-Means and Matplotlib for graphical output. The exact set of package versions installed on the project machine was not included

in the supplied source files, so this report does not invent package-version numbers or hardware performance measurements.

The implementation uses named input and output directories rather than requiring manual intervention for each processing stage. When the video identifier is S01, the common output directory contains frame images, RTMPose records, four peak workbooks and statistical results. Processing another video produces an analogous directory. The overall software can therefore operate on an arbitrary number of input files subject to practical compute and storage limits; it should not be characterized as trained or engineered specifically for the number 56.

Independent per-video outputs also mean that centroid values may differ from one recording to another. A centroid of 40 frames in one video does not necessarily correspond to the same physical repetition rate as 40 frames in another video unless their frame rates match. The current model is fitted separately for each recording rather than through a single global clustering model over all 56 videos. Cross-video comparisons should therefore first establish common units and appropriate normalization. These principles define the technical scope of the existing design and prevent misleading interpretations of its results.

CHAPTER 4
EXPERIMENTS AND RESULTS

4.1 Experimental Setup and Evidence

The developed pipeline was applied to an available collection of 56 video performances. Each video is processed independently using the same general stages and output naming convention. The dataset size describes the current experimental collection and does not constrain the program's capability to accept future recordings. A full collection-wide numerical comparison would require the output spreadsheets or summary statistics for all 56 videos.

The S01 archive supplied with this report contains the complete available case-level evidence for detailed numerical discussion. The archive includes a foot-tracking video, raw landmark coordinates, peak-detection spreadsheets for four selected landmarks, annotated peak graphs, pooled elapsed-frame data, K-Means assignments and Gaussian statistics. This chapter uses those original values and figures rather than substituting a fabricated representative example.

The purpose of this chapter is to demonstrate the analysis process, interpret its numerical outputs and describe their limitations. It does not claim an independently measured classification accuracy. The user's existing validation procedure consists of visual inspection of the observed clusters and movement graphs. The report therefore presents the speed groups as interpretable clusters, not experimentally certified ground-truth categories.

4.2 Frame and Keypoint Outputs for S01

The S01 landmark spreadsheet contains 1,788 records identified by sequential frame indices. It includes recorded image coordinates and confidence values for six foot-related landmarks. The four landmarks selected for later peak detection are the left big toe, left heel, right big toe and right heel.

The coordinate plots illustrate recurrent vertical movement rather than a single monotonic trajectory. Different landmarks can show similar broad rhythmic organization but may have unequal amplitudes and different selected minima. These differences reflect the fact that the landmarks occupy different anatomical regions and that pose estimation is performed separately for each frame. The combined visualization is useful for confirming the existence of repeated movement patterns before event reduction.

The frame count itself is not a statement about the duration of the performance without the source FPS. If the input is exactly 30 FPS, 1,788 frames correspond to approximately 59.6 seconds, but the input-file frame rate has not been independently supplied as part of the archive metadata. Accordingly, the primary reported measurements remain in frame units.

Table 4.1: Observed S01 recording and peak counts

Measurement	S01 value
Number of landmark frame records	1,788
Selected landmarks for temporal analysis	4
Left big toe detected peaks	18
Left heel detected peaks	18
Right big toe detected peaks	19
Right heel detected peaks	19
Total candidate peaks across landmarks	74
Total valid consecutive intervals	70

4.3 Detected Motion Peaks for Four Landmarks

The detected-peak graphs use blue curves to represent the original image-space Y-coordinate trajectories and red points to indicate selected local minima. Each chosen point is annotated with its frame index. The event count is not identical for all landmarks: the left big toe and left heel each have 18 selected peaks, while the right big toe and right heel each have 19. This is consistent with the independent application of the peak detector to four different signals.

Every sequence of m selected peaks yields m−1 consecutive differences. The S01 interval counts are therefore 17 for each left-foot landmark and 18 for each right-foot landmark, producing 70 pooled interval observations. This exact accounting provides an internal check of the data-loading step. It also shows why the total number of intervals differs from the total number of peak markers across the plots.

Figure 4.1 is the supplied left big toe peak plot. Corresponding figures for the other three landmarks are included to document that the procedure was performed consistently across all four signals. These plots support visual examination of event spacing, but their red markers should not be described as ground-truth contact instants without additional evidence.

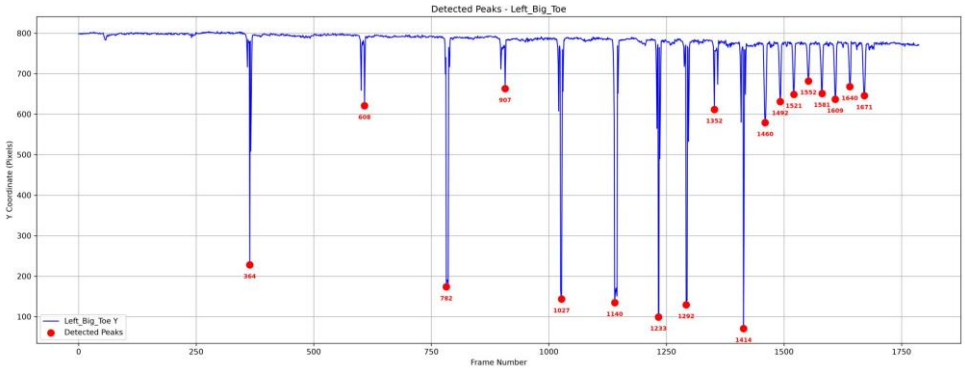


Figure 4.1: Detected peaks of the left big toe in S01

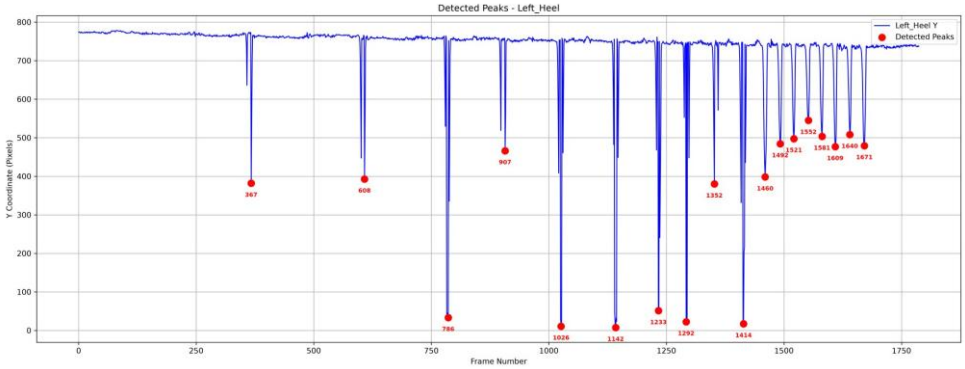


Figure 4.2: Detected peaks of the left heel in S01

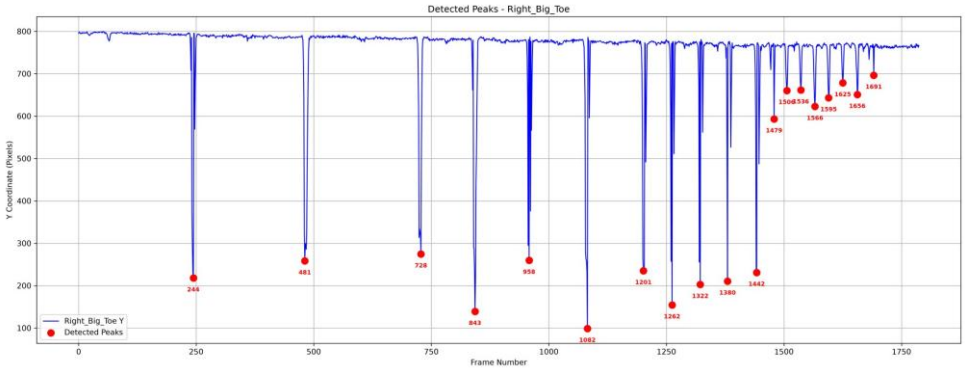


Figure 4.3: Detected peaks of the right big toe in S01

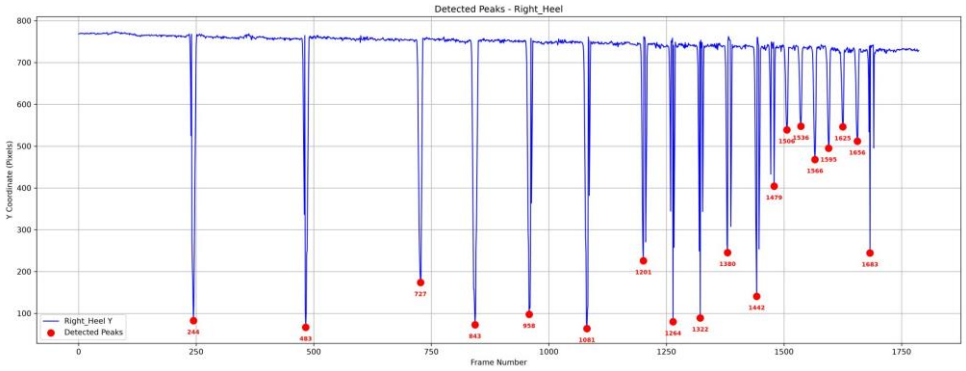


Figure 4.4: Detected peaks of the right heel in S01

Table 4.2: S01 inter-peak intervals by landmark

Landmark	Detected peaks	Computed intervals
Left big toe	18	17
Left heel	18	17
Right big toe	19	18
Right heel	19	18
TOTAL	74	70

4.4 One-Dimensional K-Means Results

The 70 valid S01 intervals are supplied together to the one-feature K-Means model. The output contains the original elapsed-frame observation, its originating landmark, the numeric cluster identifier and the mapped relative speed label. The model identifies three groups with centroid values of approximately 40.17, 115.31 and 225.50 frames. Their ordering under the implemented rules gives Fast, Medium and Slow respectively.

The Fast cluster contains 46 observations, the Medium cluster contains 16 and the Slow cluster contains 8. The unequal group sizes are an observed property of the pooled interval records; they are not automatically evidence of the relative durations of the three speed sections in the original performance. Since multiple landmarks contribute interval records, group size reflects both event detection and within-video movement frequency.

The S01 one-dimensional plot places the intervals along a horizontal axis and marks each cluster's mean with a centroid marker. Its principal result is the visible separation into groups of relatively short, intermediate and long frame intervals. A distinction must be maintained between numeric separability in the derived feature space and correct identification of the dancer's intended slow,

medium and fast sections. The latter was assessed visually in this phase, without quantitative annotation-based accuracy measurement.

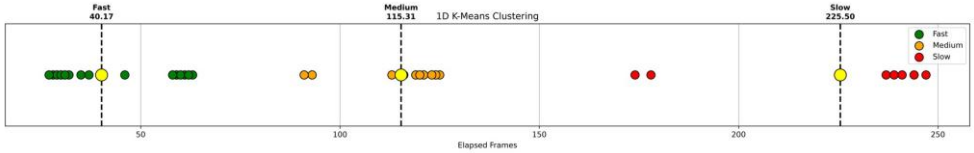


Figure 4.5: One-dimensional K-Means clusters for S01

Table 4.3: S01 cluster statistics

Speed group	Intervals (n)	Mean (frames)	SD (frames)	Observed range
Fast	46	40.17	13.99	27–63
Medium	16	115.31	9.76	91–125
Slow	8	225.50	30.73	174–247

The cluster centroids are separated by approximately 75.14 frames between Fast and Medium, and 110.19 frames between Medium and Slow. These quantities represent differences in average elapsed frames, not a controlled estimate of exact tempo ratios. The fixed distance and prominence settings affect which events enter the interval dataset, and changing their values could change the samples available to clustering.

The labeling direction is an essential part of interpretation. Lower elapsed-frame values mean that the selected events recur more frequently. Therefore, the cluster with the smallest centroid is interpreted as fastest even though the word "Fast" does not originate from a supervised target variable. The reproducible random state and repeated initializations make this numeric partition less sensitive to random initialization, but they do not resolve the need for independent speed labels if quantitative evaluation is later desired.

4.5 Gaussian Distribution Results

The Gaussian module processes the three K-Means groups independently. For S01, the estimated means match the reported cluster centroids because both calculations use the same grouped interval observations. The sample standard deviations differ between clusters: approximately 13.99 frames for Fast, 9.76 frames for Medium and 30.73 frames for Slow. As absolute frame values, these indicate the width of interval variation within the data selected into each group.

The output chart shows three normal probability-density curves constructed from the corresponding means and standard deviations. The graph also

annotates empirical coverage at one, two and three sample standard deviations. Empirical coverage is calculated from the actual data points; it is not a pre-assigned theoretical percentage. For example, the Fast group contains 32 of its 46 observations within one standard deviation of its mean, which is approximately 69.6 percent.

The Medium group has 14 of 16 observations within one standard deviation, 14 within two standard deviations, and all 16 within three standard deviations. The fact that observed two-standard-deviation coverage is 87.5 percent instead of an idealized normal percentage shows why these curves should be interpreted as descriptive fits, not as proof of a normally distributed underlying process. Cluster truncation and small sample sizes can also affect the shape of the observed distribution.

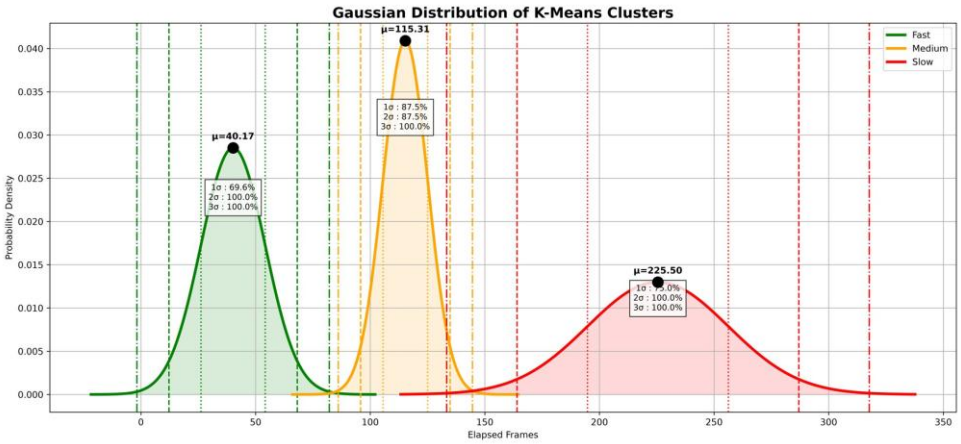


Figure 4.6: Gaussian distributions of S01 speed clusters

Table 4.4: S01 observed standard-deviation coverage

Speed group	Within ±1 SD	Within ±2 SD	Within ±3 SD
Fast (n = 46)	32 (69.57%)	46 (100%)	46 (100%)
Medium (n = 16)	14 (87.50%)	14 (87.50%)	16 (100%)
Slow (n = 8)	6 (75.00%)	8 (100%)	8 (100%)

The Gaussian statistics can be used to discuss relative temporal consistency with care. The Medium group has the smallest absolute standard deviation in this S01 result, and the Slow group the largest. However, absolute standard deviation is affected by the scale of the mean interval; a slower group naturally uses larger frame values, and its raw variation should not be read as movement complexity without normalization and independent validation.

Another methodological point is that the Gaussian curves are fitted after K-Means has already partitioned the observations based on their values. The groups are therefore selected ranges of the original interval distribution, not unconditioned random samples of independently verified speed regimes. It is reasonable to report their descriptive means, standard deviations and empirical coverage, but not to infer that the motion is intrinsically Gaussian or to claim that a given category was harder to execute.

4.6 Landmark Contributions to the Clustered Dataset

Table 4.5: S01 cluster-wise landmark contributions

Originating landmark	Fast	Medium	Slow	Total
Left Big Toe	11	4	2	17
Left Heel	11	4	2	17
Right Big Toe	12	4	2	18
Right Heel	12	4	2	18

The cluster-by-landmark cross-tabulation is obtained from the supplied K-Means spreadsheet rather than estimated from graph appearance. This view is useful because pooling the four signals can conceal how much each landmark contributes to a category. A cluster might contain observations from every selected landmark, or its membership could be influenced disproportionately by one landmark's event detector.

The presence of multiple landmarks in the merged dataset is a strength for collecting movement information from both feet, but it also complicates any attempt to count unique physical stamps. Two landmarks on one foot may respond to the same foot action. The sample-level output is correctly described as 70 landmark-specific interval observations; it should not be reported as 70 independent dance stamps or as an absolute ground-truth action count.

4.7 Interpreting Speed Dynamics from Frame Intervals

A useful derived measurement is the repetition frequency implied by an average frame interval, conditional on a known frame rate. If the source frame rate is f frames per second and the mean event interval is μ frames, the average observed event frequency is f/μ events per second. This is a conversion of units and does not establish that each detected event corresponds to exactly one musical beat or foot stamp. The present report retains cluster centers in frames because the archive does not supply a separately verified source-FPS value.

The interval sequence contains more information than three aggregate centers. Consecutive frame gaps can show gradual acceleration, abrupt changes or irregular rhythm. In the supplied S01 plots, visually decreasing event separation toward later portions is consistent with a transition from slower to faster recurrent motion. Nevertheless, because the combined K-Means input is pooled without retaining chronological order as a feature, the fitted model does not directly return temporal boundaries between speed sections.

Future analysis can link each interval back to its peak-frame pair and visualize time-local cluster labels on the original timeline. That would allow a more explicit account of when a speed transition occurred and how long a given speed section lasted. The current experimental finding is narrower but clear: the one-dimensional interval data from S01 separate into three visually inspectable ranges with distinct centroids.

$$Observed\ event\ frequency\ \approx \frac{FPS_{source}}{\mu_{interval}} \quad (4.1)$$

4.8 Evaluation Boundaries and Research Integrity

Table 4.6: What is directly verified and what remains unmeasured

Supported by existing material	Not established by existing material
Same end-to-end procedure applied to 56 videos	Aggregate mean speed across all 56 videos
Detailed S01 peak counts and interval data	Detection precision/recall against annotated foot contacts
S01 K-Means centers and cluster sizes	Ground-truth three-speed classification accuracy
S01 Gaussian statistics by cluster	Causal relation between sigma and movement difficulty
Independent per-video result directories	Generalization accuracy across dancers and cameras
Graphical qualitative examination	Statistical significance of inter-Adavu differences

Evaluation in this phase has relied on implementation outputs and visual inspection of motion and clustering graphs. Such inspection is useful for identifying gross inconsistencies or checking whether three relative interval ranges appear, but it does not substitute for an annotated test set. Precision, recall, confusion matrix and classification accuracy cannot be reported without a defined target label for each evaluated observation and a comparison between predictions and labels.

The absence of 55 additional output archives in the current report material also limits the numerical comparison that can be made across the full 56-video collection. It is correct to state that the pipeline was applied to those recordings, because that is the reported experimental scope. It would be incorrect to invent an overall centroid range, pooled variance or participant-level result. The present report therefore documents S01 in depth and treats multi-video numerical comparison as future analysis once summary outputs are consolidated.

These boundaries are productive rather than merely restrictive. They define the next steps needed to transform a working exploratory analysis into a rigorously validated movement-measurement study: event-level ground truth, FPS consistency, camera and pose quality audits, documented recording metadata and broader statistical comparison.

4.9 Detailed Numerical Interpretation of the S01 Case

4.9.1 Peak locations and temporal separation

The original peak spreadsheet for the left big toe begins with an accepted minimum at frame 364 and a second accepted minimum at frame 608. The elapsed frame difference between these observations is 244, a value included in the Slow cluster. Subsequent selected frames for this keypoint include 782 and 907, showing differences of 174 and 125 frames from their respective predecessors. These values illustrate how a simple difference operator converts frame-indexed observations into the central speed-related feature used in the study.

The frame positions are important because they connect numerical results to the actual recording. A plot of elapsed interval against the later peak frame would show whether interval length decreases gradually or in steps. The existing event plots already allow a visual assessment of this progression. By contrast, the one-dimensional K-Means scatter plot intentionally places observations according to interval magnitude, not chronological occurrence. Both visualizations are useful, but they answer different questions.

Peak counts also provide a transparent arithmetic consistency check. The two left-foot landmarks contribute 17 differences each, and the two right-foot landmarks contribute 18 differences each. Thus 17 + 17 + 18 + 18 equals 70. This total matches the number of rows in the pooled dataset and the subsequent K-Means output. The equality verifies the expected transfer of interval counts

between saved workbooks, although it does not establish that every candidate peak corresponds to a true physical action.

4.9.2 Relative interval scales and sample composition

The Fast centroid of 40.17 frames is substantially shorter than the Medium centroid of 115.31 frames and the Slow centroid of 225.50 frames. These centers are the observed means of their assigned samples. Numerically, Medium intervals average about 2.87 times the Fast mean, while Slow intervals average about 5.61 times the Fast mean. These are ratios of mean detected-event intervals, not proven ratios of the traditional performance speeds or musical beats.

The sample sizes are uneven across categories. Nearly two thirds of the pooled observations are assigned to the Fast group, while the Slow group contains only eight values. Because the Fast section of a performance contains more frequent repeated events, it may naturally generate more interval observations per unit time. Yet the exact sample count also depends on visibility, detector thresholds and contributions from the four landmarks. A larger cluster membership should therefore not be interpreted directly as a longer dance section or a more accurate category.

The observed interval ranges do not overlap across the three S01 clusters: Fast spans 27 to 63 frames, Medium spans 91 to 125 frames and Slow spans 174 to 247 frames. This separation supports the statement that the present one-dimensional partition produces distinct groups on this particular dataset. It does not imply that all performances will have equally separated ranges or that the clustering solution is optimal under every recording condition. Testing those questions requires results consolidated across the additional recordings.

4.9.3 Dispersion and the meaning of the Gaussian curves

Within the Fast cluster, the sample standard deviation is approximately 13.99 frames around a mean of 40.17 frames. The Medium group has a standard deviation of approximately 9.76 frames around 115.31 frames, while the Slow group has a standard deviation of approximately 30.73 frames around 225.50 frames. Looking only at these raw standard deviations, Medium is the narrowest cluster and Slow the widest in absolute interval units.

Mean-normalized dispersion can offer a different perspective because the scale of an interval changes substantially between the three groups. Dividing

standard deviation by mean yields approximately 34.8 percent for Fast, 8.5 percent for Medium and 13.6 percent for Slow. These derived relative percentages are based entirely on the provided S01 means and standard deviations; they do not require inventing new observations. The comparison illustrates why absolute spread and relative spread should not be confused when evaluating rhythmic timing.

The graphical Gaussian peak height is determined by the fitted standard deviation. A narrower normal curve has a higher central probability density, whereas a wider curve has a lower, broader maximum. The vertical axis is probability density, not the number of foot stamps or the probability that a particular dancer is performing correctly. The horizontal axis remains measured in elapsed frames. The shaded and dashed sigma ranges help display spread but do not independently establish the quality of the fit.

The observed one-standard-deviation coverage percentages are 69.57, 87.50 and 75.00 percent for Fast, Medium and Slow respectively. These values arise from counting actual assigned observations inside each interval. They should not be forced to match the ideal normal rule, especially because the clusters have been formed by K-Means using the same feature and the sample sizes are not large. The Slow cluster contains only eight intervals; a change in a single point's inclusion can change a coverage percentage by 12.5 percentage points. This sensitivity should be considered when using the statistics for any further interpretation.

4.9.4 What the experiment does and does not demonstrate

The S01 case demonstrates that the pipeline can transform an actual foot-motion recording into coordinate traces, a set of candidate vertical-motion events, interval measurements and three clearly separated numeric interval groups. It also demonstrates that descriptive variation can be computed and visualized independently for each group. The outputs are linked through their spreadsheet columns and file names, which supports traceability from the original model estimation to final statistical figures.

The experiment does not by itself establish the accuracy of the selected event timestamps relative to independently annotated physical foot contacts. It also does not establish that a particular Gaussian standard deviation corresponds to the subjective difficulty of Tattadavu. The appropriate empirical claim is that different relative interval groups exhibit different numerical patterns of spread.

A broader performance-complexity interpretation would require new definitions, labels and evidence.

The program was applied to 56 video performances, which establishes the current reported processing scale. For a publication-quality aggregate analysis, the per-video outputs would need to be combined into a verified summary with appropriate treatment of frame rates, repeated observations, landmark-specific differences and performer-level grouping. The absence of these aggregate files from the material available for this report means that only the detailed S01 numerical case can be independently checked here. Maintaining that distinction protects the integrity of the reported results and ensures that future extensions can build on a clearly documented baseline.

4.10 Verification Through Saved Intermediate Artifacts

An important feature of the experiment is the continuity between intermediate spreadsheet files and final plots. Each stage exposes quantities that can be checked without reproducing the entire pose-estimation procedure. The raw keypoint workbook provides coordinate observations at each frame index. The per-landmark peak workbooks specify which frames were selected as candidate events. The Elapsed_Frames column contains the difference between adjacent selected frame numbers. The merged workbook then gathers the nonmissing interval values across landmarks, and the clustering workbook appends the assigned cluster and interpreted speed label.

This series of outputs supports several internal consistency checks. First, the number of interval rows from any landmark should be one less than the number of peaks detected for that landmark, provided that every accepted peak has a valid frame index. Second, the merged row count should equal the sum of the four individual interval counts. Third, the total number of K-Means assignments should equal the number of pooled interval observations. Fourth, the number of observations reported in the Gaussian group summaries should sum to the K-Means sample count. For S01, the interval counts total 70 and the Gaussian group sizes of 46, 16 and 8 also total 70.

A further consistency check concerns cluster centers. Since the clustering algorithm assigns interval observations to groups, a reader can calculate the mean interval separately for each reported category. The S01 values obtained from the Gaussian statistics workbook match the displayed K-Means centroids to the precision shown in the plot. This agreement is expected because both

derive from the same group memberships. It is not an independent measure of classification accuracy, but it does verify that the descriptive statistics and plotting stage use the saved clustering assignments consistently.

The observed standard-deviation coverage can also be recomputed directly. For example, the Fast group contains 46 samples. Its sample mean and sample standard deviation define an interval extending one standard deviation on either side of the mean. Counting the samples within those limits produces 32 observations, corresponding to approximately 69.57 percent. The same procedure explains the coverage counts for the Medium and Slow groups. Such checks support reproducibility and make the Gaussian figure more than a decorative visualization.

These checks address integrity of the computational data flow, not external validity. A smoothly functioning sequence of file operations can still produce misleading measurements if the underlying landmark detections are wrong or if the selected extrema do not correspond to the intended dance action. For that reason, the verified saved outputs and the research limitations should be reported together. The existing S01 figures are valuable because they allow a reader to inspect both stages: the raw vertical-coordinate behavior and the final statistical representation, without assuming that the latter has already been validated against an independent reference.

CHAPTER 5
SOCIAL IMPACT AND SUSTAINABILITY

5.1 Social Impact

Bharatanatyam is transmitted through teaching, observation, repetition and embodied practice. Digital records can support this process by preserving demonstrations and providing supplementary tools for reflection. The present project offers a means of describing the timing of repeated foot motion numerically, while acknowledging that artistic achievement includes interpretation, posture, expressiveness and context that cannot be reduced to one speed feature.

The most immediate educational value lies in interpretability. Teachers or learners can inspect where the landmark trajectory oscillates, which candidate motion events were detected and how their intervals differ. Such visual and numeric evidence can support discussion of rhythmic practice. The method should be presented as an analytical aid rather than as an autonomous authority on correct Bharatanatyam technique or a substitute for expert instruction.

The project also illustrates a potential route toward preservation of intangible cultural movement patterns. Structured datasets organized by Adavu and performance can make recorded examples easier to study and compare. A repeatable processing format enables researchers to investigate temporal organization over time, subject to appropriate permissions for collecting, storing and sharing recordings.

Responsible deployment would require attention to dancers' privacy, consent and intellectual or cultural context. Video is potentially identifiable, even when the analysis concentrates on foot movement. Any future dataset distribution should specify permitted uses and avoid exposing recordings without authorization. Technical results must be communicated with adequate uncertainty, particularly when the system has not yet been independently validated against expert annotations.

5.2 Sustainability

The current workflow relies primarily on software-based image analysis and statistical processing rather than external force plates or wearable devices. That reduces dependence on specialized instrumentation when recordings

already exist. It also provides an opportunity to reuse archival or classroom video, where consent and technical quality permit, instead of arranging a new sensor-based capture session for every analysis.

The modular program can process variable numbers of videos by iterating through available MP4 files. Individual per-video output directories support reinspection and selective rerunning of stages. Persistent intermediate files reduce the need to repeat the entire analysis when only a downstream plot or statistical summary must be examined.

There are also computational costs. Pose estimation can require substantial processing time and storage because every frame is analyzed and the program saves individual JPEG frames, spreadsheets, graphs and an annotated video. CPU inference and parallel worker execution affect resource use. A future implementation might use more efficient intermediate representations, adjustable output retention, validated batch scheduling and hardware selection suited to the available computational environment.

Long-term maintenance will benefit from documented library versions, consistent filename conventions, stored configuration parameters and preservation of the source FPS for each recording. These measures would improve reproducibility and prevent ambiguity when recordings come from different devices or are processed again at a later date.

5.3 Ethical and Practical Considerations

Any automated analysis of a culturally grounded art form must distinguish a measurable signal from the broader quality of performance. A smaller inter-event interval describes more frequent selected events; it does not by itself imply better execution. Similarly, a small within-cluster standard deviation describes tighter timing variation but not necessarily artistic appropriateness or low physical difficulty.

Future evaluation should involve domain experts when defining intended speed sections, foot-action events and acceptable timing variations. Collaboration with trained dancers would help determine which visual event best corresponds to a Tattadavu stamp and whether the intended traditional categories are faithfully represented by the computed clusters. Such involvement can improve both technical validity and respectful interpretation of the recorded performances.

5.4 Potential Educational Workflow and Limits of Automated Feedback

A possible future educational workflow could begin with a teacher choosing a representative Tattadavu recording and marking the intended speed sections. A learner's recording could be processed using the same keypoint extraction pipeline, followed by an inspection of its event trajectories and timing intervals. The resulting graphs would offer a focused starting point for discussing whether the observed movement frequency increases as intended and whether individual repeated actions are unusually separated in time.

This workflow would still require human interpretation. A pose-estimation system does not automatically recognize whether a stance satisfies the dancer's training tradition, whether movement amplitude is intentional or whether rhythmic emphasis matches the artistic context. In particular, a learner who produces perfectly regular but incorrectly placed movements could show low interval variance while still performing the Adavu inadequately. Quantitative regularity is one dimension of performance, not a complete score.

Accessibility is another consideration. Video-based analysis may be useful when specialized biomechanics equipment is unavailable, but it is not free from technical constraints. Adequate framing, image quality, consistent camera setup and sufficient computer resources may be required. Any software interface intended for broad educational use should communicate when pose confidence is low or when the detected events cannot be trusted. It should also avoid presenting probabilistic or clustering results as unquestionable expert judgments.

For cultural preservation, the same system could support a searchable archive where each authorized recording retains its metadata, movement coordinates and temporal summaries. Responsible stewardship would include clear permissions for storage, attribution and research reuse. Such an archive would complement human teaching and documentation, rather than reducing a complex artistic tradition to a handful of statistical measurements.

CHAPTER 6
CONCLUSION

The Phase I project developed and applied a modular framework for studying speed dynamics in Bharatanatyam Adavus using foot movement extracted from ordinary video. The completed system reads MP4 recordings, extracts video frames, applies RTMPose WholeBody landmark estimation, creates vertical-coordinate trajectories for the relevant foot landmarks, identifies recurring candidate motion events and calculates the intervals separating them. The resulting one-dimensional elapsed-frame observations are clustered using K-Means, and their dispersion is examined using Gaussian descriptive statistics.

The design is independent of a fixed dataset size. The user's available experimental collection comprised 56 video performances, all processed using the same procedure and a consistent output structure. This demonstrates the practical use of the pipeline over a multi-video collection, while the current report contains detailed original measurements only for the supplied S01 experiment.

For S01, the available files cont7ain 1,788 frame records and 74 selected peak markers across four landmarks, yielding 70 valid consecutive intervals. K-Means separated these values into three relative categories with average intervals of 40.17, 115.31 and 225.50 frames for Fast, Medium and Slow. The Gaussian module produced separate mean, variance, standard deviation and empirical coverage summaries, allowing the dispersion within each group to be described numerically. These are directly supported results of the implemented software rather than hypothetical example values.

The main achievement is an interpretable end-to-end transformation from foot-motion video to candidate event timing, speed-related clusters and statistical summaries. Limitations remain: coordinate minima have not been independently proven to equal exact floor contacts, true speed-category accuracy has not been measured, and cluster-specific spread has not been validated as a measure of difficulty. Recognizing these boundaries keeps the report faithful to the present stage of research and defines a clear path toward more rigorous analysis.

Overall, the work provides a reproducible foundation for quantitative study of rhythmic foot movement in Bharatanatyam. It can be extended to additional

recordings and Adavus while preserving the same core strategy of pose-driven temporal feature extraction and interpretable analysis.

CHAPTER 7
FUTURE SCOPE

7.1 Expansion to Eight Additional Bharatanatyam Adavus

The primary objective of future work is to extend the existing speed dynamics analysis framework from Tattadavu to eight additional Bharatanatyam Adavus. Since different Adavus involve distinct foot movements and rhythmic patterns, the existing keypoint tracking, peak detection, K-Means clustering and Gaussian analysis techniques will be adapted and evaluated accordingly. This extension will enable comparative analysis of speed variations and rhythmic consistency across multiple Adavu categories.

7.2 Improvement in Speed Classification and Validation

Future work will focus on improving the accuracy and reliability of speed classification. The detected movement events and their corresponding slow, medium and fast speed categories will be validated against expert-annotated performances. Evaluation metrics such as accuracy, precision, recall and F1-score can be used to assess classification performance. Additionally, converting frame intervals into time-based measurements will enable consistent analysis across videos recorded at different frame rates.

7.3 Temporal Analysis and Performance Comparison

The existing framework will be enhanced to identify speed transitions within individual performances and analyze variations in rhythmic execution over time. Techniques such as temporal segmentation and Dynamic Time Warping (DTW) may be explored to compare corresponding movement sequences across different performances. This will help investigate timing consistency and movement variations between dancers and Adavu categories.

7.4 Enhancement of the User Interface and Practical Applications

The existing video-upload and processing interface will be enhanced to support the additional Adavus and provide improved visualization of foot trajectories, detected movement events, speed classifications and statistical results. The long-term objective is to develop an accessible analysis tool that can assist Bharatanatyam dancers, instructors and researchers in understanding rhythmic execution and performance consistency.

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APPENDIX-1

REPRESENTATIVE OUTPUT STRUCTURE AND VERIFICATION

The following file layout describes the original S01 output archive reviewed for the present report. It is supplied to document reproducibility and does not imply that only one video can be processed. The same general output structure was used for the other recordings in the locally stored experimental set.

S01/RTM_Detected_Frames/foot_keypoints.xlsx
S01/RTM_Detected_Frames/foot_tracking.mp4
S01/RTM_Detected_Frames/graphs/*.png
S01/output_1/detected_peaks_Left_Big_Toe.xlsx
S01/output_1/detected_peaks_Left_Heel.xlsx
S01/output_1/detected_peaks_Right_Big_Toe.xlsx
S01/output_1/detected_peaks_Right_Heel.xlsx
S01/output_1/detected_peaks_*.png
S01/Statistics_Output/merged_elapsed_frames.xlsx
S01/Statistics_Output/kmeans_results.xlsx
S01/Statistics_Output/gaussian_statistics.xlsx
S01/Statistics_Output/04_1D_KMeans.png
S01/Statistics_Output/05_Gaussian.png

Unrelated analysis artifacts that coexist in the original archive are not used in this project report.